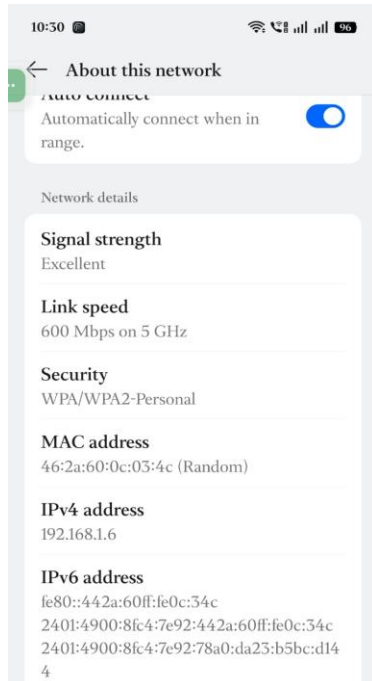


1) Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.
Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.



```
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ITCPL>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:
Reply from 192.168.1.6: bytes=32 time=61ms TTL=64
Reply from 192.168.1.6: bytes=32 time=6ms TTL=64
Reply from 192.168.1.6: bytes=32 time=4ms TTL=64
Reply from 192.168.1.6: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 61ms, Average = 18ms

C:\Users\ITCPL>
```

```
10:34
- Install: pkg install <package>
Upgrade: pkg upgrade
Report issues at https://bugs.termux.com

$
$ ping 192.168.1.16
PING 192.168.1.16 (192.168.1.16) 56(84) bytes of data:
64 bytes from 192.168.1.16: icmp_seq=1 ttl=128 time=4.57 ms
64 bytes from 192.168.1.16: icmp_seq=2 ttl=128 time=4.43 ms
64 bytes from 192.168.1.16: icmp_seq=3 ttl=128 time=18.7 ms
64 bytes from 192.168.1.16: icmp_seq=4 ttl=128 time=14.3 ms
64 bytes from 192.168.1.16: icmp_seq=5 ttl=128 time=19.0 ms
64 bytes from 192.168.1.16: icmp_seq=6 ttl=128 time=5.73 ms
64 bytes from 192.168.1.16: icmp_seq=7 ttl=128 time=4.31 ms
64 bytes from 192.168.1.16: icmp_seq=8 ttl=128 time=4.18 ms
64 bytes from 192.168.1.16: icmp_seq=9 ttl=128 time=15.7 ms
64 bytes from 192.168.1.16: icmp_seq=10 ttl=128 time=16.6 ms
64 bytes from 192.168.1.16: icmp_seq=11 ttl=128 time=5.77 ms

ESC / — HOM ↑ END PGUP
TAB CTRL ALT ← ↓ → PGD
```

2) Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
C:\WINDOWS\system32\cmd. X + v

Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ITCPL>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482a:400::]
over a maximum of 30 hops:

 1      *          *          *      Request timed out.
 2     2 ms       2 ms       3 ms    undefined.hostname.localhost [2402:a00:404:1::1111]
 3     3 ms       3 ms       3 ms    undefined.hostname.localhost [2402:a00:1::3d]
 4    28 ms      28 ms      28 ms    undefined.hostname.localhost [2402:a00::a]
 5    28 ms      28 ms      28 ms    2001:4860:482a:400::

Trace complete.
```

3) Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ITCPL>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.1.16: Destination host unreachable.
Reply from 192.168.1.17: Destination host unreachable.

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\ITCPL>
```

4) Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```
C:\WINDOWS\system32\cmd. x + v
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\ITCPL>arp -a

Interface: 192.168.1.17 --- 0x4
Internet Address      Physical Address      Type
192.168.1.1           28-77-77-6b-8f-84    dynamic
192.168.1.4           04-7c-16-d3-5c-58    dynamic
192.168.1.8           00-4e-01-ba-4b-a3    dynamic
192.168.1.23          e4-a8-df-da-57-80    dynamic
192.168.1.30          8c-04-ba-9d-d6-8b    dynamic
192.168.1.35          5c-3a-45-38-54-fb    dynamic
192.168.1.36          e4-b9-7a-eb-b3-a4    dynamic
192.168.1.46          50-03-cf-1a-87-07    dynamic
192.168.1.66          20-2b-20-68-12-47    dynamic
192.168.1.213         28-18-fd-71-fa-33    dynamic
192.168.1.214         28-18-fd-71-fa-28    dynamic
192.168.1.215         28-18-fd-71-fa-45    dynamic
192.168.1.216         28-18-fd-71-f4-36    dynamic
192.168.1.217         28-18-fd-59-37-bf    dynamic
192.168.1.218         28-18-fd-59-37-d5    dynamic
192.168.1.219         28-18-fd-59-37-b4    dynamic
192.168.1.220         28-18-fd-59-37-c3    dynamic
192.168.1.221         28-18-fd-59-37-b5    dynamic
192.168.1.222         28-18-fd-5d-79-1e    dynamic
192.168.1.245         28-18-fd-42-1e-dc    dynamic
192.168.1.249         6c-3c-7c-19-e4-2f    dynamic
192.168.1.250         00-20-6b-f0-16-84    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22           01-00-5e-00-00-16    static
224.0.0.251          01-00-5e-00-00-fb    static
224.0.0.252          01-00-5e-00-00-fc    static
239.255.255.250      01-00-5e-7f-ff-fa    static
255.255.255.255      ff-ff-ff-ff-ff-ff    static

Interface: 192.168.56.1 --- 0x10
Internet Address      Physical Address      Type
192.168.56.255        ff-ff-ff-ff-ff-ff    static
224.0.0.2             01-00-5e-00-00-02    static
224.0.0.22           01-00-5e-00-00-16    static
224.0.0.251          01-00-5e-00-00-fb    static
224.0.0.252          01-00-5e-00-00-fc    static
```

5) Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

Step 1: Connecting the Devices

First, I connected my windows laptop and my friend's Android smartphone to the same Wi-Fi network. I checked that both devices were connected to the same network and received IP addresses from the Wi-Fi router.

Step 2: Checking the IP Address

On the Windows laptop, I opened Command Prompt and entered:

```
ipconfig
```

The laptop received an IP address such as 192.168.1.101. I also checked the smartphone's Wi-Fi settings and found its IP address, for example 192.168.1.102.

Step 3: Testing Connectivity Using Ping

I tested communication between the laptop and smartphone by entering the following command:

```
ping 192.168.1.102
```

The result showed replies from the smartphone:

```
Reply from 192.168.1.102: bytes=32 time=5ms TTL=64
```

```
Reply from 192.168.1.102: bytes=32 time=4ms TTL=64
```

```
Reply from 192.168.1.102: bytes=32 time=6ms TTL=64
```

This indicated that the laptop could reach the smartphone over the Wi-Fi network.

Step 4: Testing Using Traceroute

Next, I used the Windows tracert command to examine the network path to a website:

```
tracert google.com
```

The command displayed several network hops between my laptop and the destination server. The first hop was normally the local Wi-Fi router, followed by other routers operated by the Internet Service Provider (ISP).

Step 5: Checking the ARP Table

Finally, I used:

arp -a

This displayed IP addresses and corresponding MAC addresses stored in the laptop's ARP table. I used these details to identify devices that had recently communicated with the laptop.

Results and Issues Observed

The ping test was successful, showing that the laptop and smartphone could communicate on the local Wi-Fi network. The tracert test successfully displayed the route to the Internet destination, although some hops could show Request timed out because routers sometimes block traceroute responses. The arp -a command also displayed local IP and MAC address information.

Conclusion

The practical successfully demonstrated basic network configuration and connectivity testing. Using ipconfig, ping, tracert, and arp -a helped me understand IP addressing, local device communication, routing, and MAC-address discovery on a Wi-Fi network.